NDSEG Application

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1 Personal Essay Question 1

No more than 500 words and no special formatting required for each essay. Answer the following: What are your short and long-term professional goals? How did these goals develop? How have you already begun to lay the foundation for these goals? How does this fellowship fit into these goals?

My short-term professional goals center on becoming a full-time researcher at a national laboratory, where I can apply my skills at the intersection of mechanical engineering and machine learning. I aim to contribute to pioneering research at national laboratories advancing nuclear energy or strengthening energy infrastructure. Specifically, I'm interested in developing data-driven models for energy system optimization and autonomous mechanical systems. Through the rest of my graduate education, I aim to deepen my expertise in scientific machine learning, focusing on developing models that are able to better analyze, identify, and control dynamics for complex mechanical systems.

In the long term, I aspire to lead a research group dedicated to addressing the nation's most pressing scientific and technological challenges. I currently view these challenges as strengthening the security, resilience, and accessibility of nuclear and other nationally significant energy sources. Whether the future demands breakthroughs in energy, grid resilience, or defense technologies, I want to direct research efforts toward these national priorities. Through this work, I strive to leave a tangible, positive impact on the nation's technological and infrastructural landscape.

These goals developed as I progressed through my education. I enjoyed mechanical engineering from the beginning, but internships at local companies left me feeling unsatisfied. I realized that I was motivated not just by engineering, but by the opportunity to work on problems with national and global importance. Motivated by this insight, I pursued graduate education, which I hoped would open the path to research roles tackling these challenges. During my master's research on point cloud filtration, I realized my passion for designing experiments, testing new methods, and contributing knowledge that directly informs applied engineering solutions. These experiences solidified my conviction to run a laboratory, creating an environment where others can collaborate to push the boundaries of both science and application.

I have already begun to lay the foundation for these goals. Over the past three years, I have worked as a graduate research intern at a national lab, where I developed a computer vision model for detecting mechanical part anomalies, contributing directly to national energy and infrastructure priorities. Simultaneously, I have pursued my PhD, focusing on developing scientific machine learning methods with applications to energy and infrastructure challenges. These experiences strengthened my ability to design novel machine learning methods, apply them to real-world energy problems, and publish results for peer review.

This fellowship is a critical step in advancing my goals. In the short term, the fellowship will provide the resources to advance my research on machine learning-enhanced control systems, allowing me to focus on high-impact projects directly aligned with my goal of improving autonomous system performance. It will also expand my network within federal research and provide valuable insight into how federally funded science is conducted and prioritized. In the long term, the fellowship will help establish me as a scientist prepared to lead research efforts that serve the nation and contribute to solving its most important challenges.

2 Personal Essay Question 2

No more than 500 words and no special formatting required for each essay. Personal story statement to describe the Fellows personal journey to get where they are today (e.g., 1st to receive PHD, 1st to go to college, worked multiple position while achieving degree, what it took personally to get to where they are, etc. . .)

Coming from a family with no history of advanced degrees, I never imagined I would be pursuing a Ph.D. My sister and I are the first in our extended family to take this path, and together we've navigated the uncertainty of graduate school, including finding mentors, learning to do research, and pushing through challenges without a roadmap. Along the way, I've been fortunate to have guides who believed in me, especially the professor who first introduced me to research and later became my master's advisor, who taught me how to design experiments, persevere as a scholar, and navigate graduate school's challenges.

Of these challenges, finances was a major one. Because I was confident I wanted to pursue graduate school I worked internships every summer to help cover tuition. I also began work as an undergraduate research assistant, which gave me both financial support and hands-on experience. These roles developed my confidence, skills, and built the foundation for the career I wanted.

What began as an undergraduate research assistant role grew into a graduate research assistantship during my master's degree. One defining moment from that time was publishing my first journal article. Sharing my work with the broader scientific community gave me a feeling I had never experienced before, a feeling that I could contribute to something larger than myself. That experience not only strengthened my resolve to pursue a Ph.D., but also sparked my long-term goal of leading a research lab, to guide others toward scientific discovery.

During graduate school, I have nurtured my passion for guiding others in scientific discovery by mentoring a high school robotics team. Throughout my master's degree, I devoted fifteen hours a week to this work while completing my own research and taking classes. Balancing these priorities was exhausting, and at times I considered stepping back. But seeing a student's excitement as they debugged a robot at a competition reminded me why I had chosen this path, and it sustained me through the most demanding weeks. As I continue my Ph.D., I am committed to remaining active in the robotics community and fostering the same sense of discovery in others that drives my own research.

Much of the resilience that enabled me this far comes from my family, who instilled in me the importance of curiosity, education, and perseverance. The mindset of "the job's not done till it's done" carried me through the hardest points in my education and continues to guide me today. I'm proud of how far my sister and I have come, navigating much of this journey ourselves with guidance from mentors. What motivates me now is the thought that one day my work could directly help solve problems that improve people's lives. This sense of purpose, to apply my skills to solving real-world problems, continues to drive me forward, shaping the scientist and mentor I aspire to be.

3 Proposal Abstract

This fellowship application proposes the development of a Parameterized Laplace Multi-Neural Operator (PLMNO) enabled Model Predictive Control (MPC) algorithm. The PLMNO is a novel machine learning-enabled neural operator designed to efficiently solve multiple types of differential equations across an infinite range of parameter configurations, enabling rapid and accurate modeling of complex dynamical systems. By integrating the PLMNO into an MPC framework, the proposed research aims to significantly reduce the computational burden typically associated with traditional finite difference or numerical solvers for MPC while improving robustness and fidelity when compared with traditional MPC implementations. The PLMNO-enabled MPC system will be evaluated on a variety of synthetic and real-world canonical nonlinear benchmark systems, including both chaotic and non-chaotic dynamics, to assess predictive accuracy, robustness, and generalization. The proposed PLMNO-enabled MPC algorithm advances the fundamental mathematics of operator-based control for nonlinear, stochastic, and adversarial nonlinear systems, including rigorous analysis of stability, convergence, and robustness properties across parameterized families of differential equations. Ultimately, this research supports the Air Force's and Army's goals for next-generation autonomous systems by delivering a computationally efficient, resilient, and affordable control framework that can withstand adversarial conditions and operate effectively in the field without relying on heavy onboard computing resources.

Parameterized Laplace Multi-Neural Operator Enabled Model Predictive Control

1 Introduction and Background

Model Predictive Control (MPC) is a control strategy that optimizes system behavior over a finite time horizon by predicting future states and computing control actions that minimize a cost function while enforcing constraints [7]. MPC has become a cornerstone in autonomous systems, particularly drones, where it enables precise flight stabilization, trajectory tracking, obstacle avoidance, and adaptive response to changing conditions [14]. In addition to enabling highly reliable control, MPC can complement reinforcement learning (RL) based strategies to provide a robust system without the immense training and computational costs of RL alone [16]. In military applications, autonomous vehicles must operate under highly uncertain and unpredictable environments, where factors such as wind gusts, turbulence, or sudden payload changes can significantly affect system performance. Improvements in MPC algorithms directly enhance the adaptability, resilience, and efficiency of drones and other autonomous platforms [2], which aligns with the AFOSR Dynamical Systems and Control Theory program (BAA Sec. A.2.c), emphasizing rigorous methods for control of nonlinear, uncertain, and adversarial systems [1].

Traditional MPC implementations can be computationally intensive [15]. Their complexity limits real-time applicability and feasibility for lightweight, low-cost platforms [9]. As system nonlinearity, environmental variability, or real-time requirements increase, traditional methods often fail to meet operational constraints [5]. Neural operators offer a promising solution, providing data-driven approximations of differential equation solutions [10]. These models learn mappings between functions, allowing rapid prediction of system dynamics without explicitly solving underlying equations. The Laplace Neural Operator (LNO) is particularly effective [3], transforming differential equations into the Laplace domain to capture complex behaviors using relatively few trainable parameters. This efficiency makes the LNO well-suited for embedded or low-power platforms, enabling onboard predictive control for drones and other vehicles that operate with limited computational resources.

Neural operator-enabled MPC can achieve high predictive accuracy while dramatically reducing computational requirements [6]. This capability is critical for drones and other autonomous vehicles, where strict weight, power, and cost constraints make traditional MPC challenging. Moreover, the data-driven approach allows modeling of systems with unknown or complex governing equations [8], enabling predictive control in situations where first-principles modeling is impractical. However, current neural operators have key limitations. Most are trained for a single configuration of system parameters [12], so any changes in mass, stiffness, or external conditions that are not captured by the dynamic system's forcing function i.e., excitation signal, necessitate retraining. This means that sensor data such as wind speed, temperature, and humidity often cannot be incorporated into a neural operator's prediction. Additionally, operators are usually restricted to a single type of differential equation. These limitations reduce applicability in variable, real-world operational environments.

To address these challenges, this proposal develops the Parameterized Laplace Multi-Neural Operator (PLMNO) enabled MPC framework. The PLMNO-enabled MPC allows autonomous vehicles to adapt to changing conditions without retraining, operate effectively in sensor-rich, dynamically varying environments, and run efficiently on low-cost hardware. By combining computational efficiency with broad applicability, this research advances robust, agile, and cost-effective autonomous control, directly supporting real-world deployment in unpredictable or adversarial conditions [4].

This research aligns directly with NDSEG objectives by providing advanced training in high-impact, military-relevant science and engineering disciplines. Developing the PLMNO-enabled MPC framework integrates applied mathematics, control theory, and machine learning, equipping the applicant with expertise in predictive modeling and real-time control of nonlinear, uncertain, and hybrid systems. The project also fosters long-term relationships with DoD research sponsors through direct engagement with Air Force and Army applications, supporting mentoring and collaboration opportunities consistent with NDSEG goals. Finally, by applying cutting-edge techniques to defense-relevant autonomous systems, the work facilitates the applicant in pursuing a PhD focused on DoD mission-related research, contributing to the training of the next generation of U.S. citizens in science and engineering of military importance.

2 Primary Objectives

The proposed work focuses on three primary objectives as outlined:

1. Develop and evaluate the Parameterized Laplace Multi-Neural Operator (PLMNO) using synthetic data of 1D and 2D system, including chaotic and non-chaotic dynamics.
2. Develop and evaluate a PLMNO enabled Model Predictive Control (MPC) framework using synthetic environments to assess this framework's computational efficiency, data efficiency, and robustness versus traditional MPC methods.
3. Evaluate a PLMNO enabled MPC framework on synthetic and real-world traditional controls tasks that include changing environmental and physical variables like wind speed, mass, and moment of inertia.

This work will be conducted under Dr. Guang Lin at Purdue University, with access to extensive computational resources and faculty expertise beyond my advisor's focus in scientific machine learning.

2.1 Research Objective 1

Developing and evaluating the Parameterized Laplace Multi-Neural Operator (PLMNO), will first involve the design and creation of the overall neural network's architecture. Shown in Figure 1 is the proposed architecture for the PLMNO, where $f(t)$ represents an input forcing function, ID represents an identifier to inform the PLMNO which dynamic system is to be solved (e.g., Duffing oscillator, driven pendulum, etc.), P represents a vector of physical parameters relevant to the system, $\mathscr{L}$ represents the Laplace transform, $\oplus$ represents concatenation, and W_1, W_2, W_3, and W_4 are learnable parameters of the neural network. Further details regarding the mechanisms of the LNO can be found in [3].

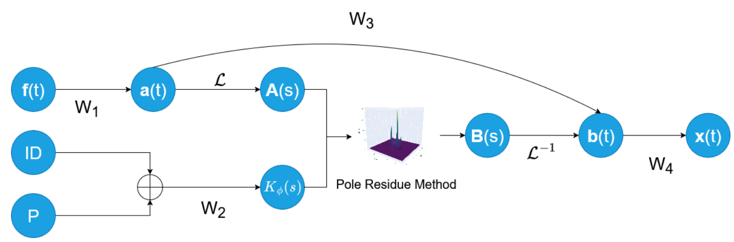

Figure 1: The proposed architecture of the PLMNO.

To rigorously assess the performance of the PLMNO, a single instance of the model will be trained and tested on a wide variety of synthetic i.e., simulated, datasets representing both 1D and 2D systems. These datasets will include canonical chaotic and non-chaotic dynamics, which can be extended to hybrid and multi-scale systems relevant to complex autonomous vehicles. By incorporating a broad spectrum of behaviors, the PLMNO will be evaluated not only on its ability to reproduce stable dynamics but also on its capacity to model systems that are highly sensitive to initial conditions. This approach ensures that the PLMNO is capable of generalizing across system types and complexity levels, a critical requirement for real-world deployment in autonomous control.

2.2 Research Objective 2

The next research objective focuses on the integration of the PLMNO into an MPC framework. This integration will replace traditional numerical methods that an MPC would normally use with a trained PLMNO. To evaluate this PLMNO-enabled MPC framework, experiments will be conducted using a wide variety of synthetic environments designed to mimic realistic control systems that an MPC algorithm might be used on. These environments will include stabilizing an inverse pendulum and controlling a Duffing oscillator. Additionally, ablation studies will be performed to assess how the PLMNO-enabled MPC performs under restrictions like less available training data and reduced floating point accuracy. This ablation study will indicate how realistic this PLMNO-enabled MPC architecture is in the real world where training data is expensive to generate, and how suitable it is for large scale applications, like complex multi-agent control systems. The goal of this ablation study will be to maintain optimal performance levels down to 4-bit floating points, a very low level of precision, for the weights and biases of the PLMNO, a goal based on findings from similar past research [13]. Comparative analyses against traditional MPC will assess computational efficiency, accuracy, and robustness. Demonstrating superior performance in synthetic settings will establish PLMNO-enabled MPC as a viable next-generation control strategy, paving the way for its application in real-world, hybrid, non-equilibrium systems.

2.3 Research Objective 3

The final objective extends evaluation beyond controlled simulated testing conditions into more realistic ones. The PLMNO-enabled MPC framework will be applied to real-world control tasks where system parameters and environmental conditions vary over time. These scenarios will incorporate stochastic perturbations and adversarial disturbances, including changing wind fields, varying mass/inertia, and air turbulence. By introducing stochastic and adversarial perturbations, this objective will rigorously evaluate the PLMNO-enabled MPC's stability, convergence, and robustness across non-stationary system dynamics. The evaluation will occur under conditions that closely mirror the operational challenges faced by drones and other autonomous vehicles in adversarial environments, such as explosion-induced shockwaves and rotor turbulence from nearby vehicles like helicopters.

The experimental design will compare PLMNO-enabled MPC against traditional MPC methods in these environments, as well as MPC-enabled RL methods, with a focus on quantifying improvement to robustness against disturbances, training data required, and computational demands on embedded hardware. Real-world tests will include classic control problems including an inverted pendulum, as well systems specifically relevant to the Air Force and Army, like the constant-thrust regulation of a rotor subject to a time-varying moment of inertia or turbulent air conditions. This objective aims to validate the PLMNO-enabled MPC framework as a practical, lightweight, and scalable solution for real-world autonomous control applications. The goals for this validation are a 10-fold improvement on computation time when compared with traditional MPC algorithms, and a similar improvement on required training data when compared with a traditional RL approach, similar to improvements that have been shown in past related research [11], [17].

3 Preliminary Results

Initial development of the Parameterized Laplace Multi-Neural Operator (PLMNO) has been completed, and its performance has been evaluated on a set of canonical 1D dynamical systems. Early testing has focused on the Duffing oscillator, a forced pendulum, and a Lorenz system. In these experiments, the PLMNO demonstrated the ability to accurately reproduce system trajectories across a wide range of system parameters and forcing inputs with small deviations from the ground truth. Notably, a single trained model was able to capture the behavior of multiple nonlinear systems, including one with chaotic behavior. Shown below, in Figures 2, 3, and 4 are three instances of a single trained PLMNO model being used to predict the output of these dynamic systems when provided the respective forcing functions.

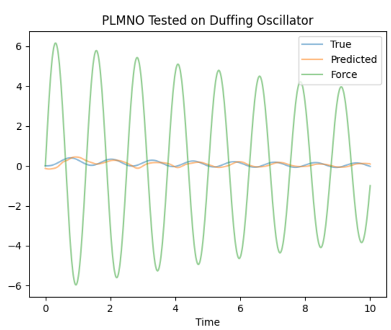

Figure 2: PLMNO tested on Duffing oscillator

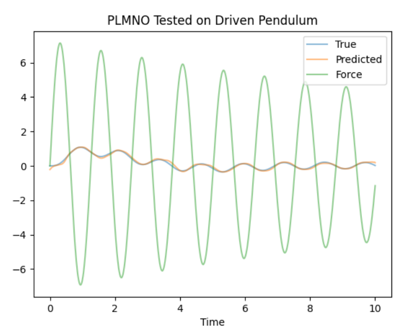

Figure 3: PLMNO tested on forced pendulum

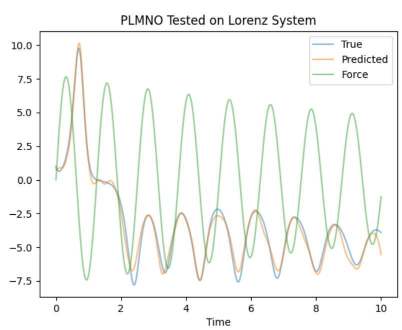

Figure 4: PLMNO tested on Lorenz system

4 Conclusion

This research will advance neural operator-enabled MPC frameworks, enabling rapid and accurate modeling of complex system dynamics under changing conditions. For the U.S. Air Force, this capability translates into more resilient and adaptive drone platforms for contested and uncertain environments. A potential deployment roadmap progresses from simulation, to hardware-in-the-loop testing, and ultimately to flight testing; full-scale flight tests are left to future work to keep the current proposal feasible. This proposal focuses on the first two stages, providing a practical foundation for validating the PLMNO-enabled MPC approach.

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